Original Mixed-radix: Final Evidence and Closure

A real mechanism, but no stable natural-data method

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1. The Answer First

Final result for the tested method

The original fixed-adjacent mixed-radix method is **not competitive on SIFT1M or GIST1M** under the frozen Recall–speed evaluation.

- At 32 bytes, it becomes exactly the power-of-two control.
- At 64 bytes, Recall changes by at most about one tenth of one percentage point, with both positive and negative signs.
- A synthetic positive control now confirms that the intended mechanism itself is real and reaches the query result.

Interpretation

The implementation can exploit non-power-of-two cardinalities. The tested natural data simply does not contain enough of this structure for a useful win.

2. Scope Correction: Which “Attempt 4” Is This?

Original target: this deck

Two adjacent scalar quantizers share one packed word. Their numbers of levels may be arbitrary positive integers.

Separate branch: not this result

The structured-2D method learns a reusable two-dimensional shape with affine transforms. It was evaluated separately and is currently set aside.

Why the distinction matters

Mixed-radix changes how a fixed number of bit patterns is divided between two scalar coordinates. It does **not** learn a shared 2D vector codebook.

3. The Original Intuition

Four bits give exactly $2^4 = 16$ possible labels for one coordinate pair.

Power-of-two allocation

Each coordinate receives 1, 2, 4, 8, ... levels. A typical balanced split is $4 \times 4 = 16$.

Mixed-radix allocation

Allow any integers K_1, K_2 as long as

$$K_1 K_2 \leq 16.$$

This includes $3 \times 5 = 15$.

Hypothesis

If one coordinate really needs 3 values and the other needs 5, mixed-radix can retain all 15 combinations. Separate binary rounding would need $4 \times 8 = 32$ states and cannot fit in four bits.

4. Running Example: Pack a 3×5 Grid

Let $z_1 \in \{0, 1, 2\}$ and $z_2 \in \{0, 1, 2, 3, 4\}$. Store

$$\text{label} = z_1 + 3z_2.$$

	$z_2 = 0$	1	2	3	4
$z_1 = 0$	0	3	6	9	12
$z_1 = 1$	1	4	7	10	13
$z_1 = 2$	2	5	8	11	14

Every real combination has a unique label; label 15 is simply unused.

What the power-of-two control must do

With only 16 labels it may choose 4×4, but then five distinct values of the second coordinate must share only four reconstructed values. At least two real prototypes can receive the same stored code.

5. Training, Encoding, and Querying

```
Training, using base/index data only:
  for each adjacent coordinate pair:
    estimate scalar reconstruction cost for each level count
    choose K1, K2 minimizing total cost, subject to K1*K2 <= 2^B

Encoding one database vector:
  z1 = nearest scalar centre on coordinate 1
  z2 = nearest scalar centre on coordinate 2
  store label = z1 + K1*z2

Querying:
  build the complete 2^B-entry distance table for each pair
  decode one label per pair, add table values, keep the best 100 IDs
```

Controlled difference

Arbitrary arm A and power-of-two arm D use the same pairs, bytes, candidates, table builder, and scan loop. Only the allowed values of K_1, K_2 differ.

6. What We Actually Compared

A128	arbitrary integer radices, such as $15 \times 17 = 255$
D128 FULL	identical method, but each radix must be a power of two
PQ / OPQ	standard product-quantization baselines for deployability context
data	SIFT1M and GIST1M
payload	exactly 32 or 64 code bytes per database vector
index sizes	1,024 and 4,096 coarse database cells
quality	Recall@100: how many of the true nearest 100 are returned
speed	single-query latency and 12-core queries per second
search range	nine fixed numbers of cells searched per query

Fairness rule

A and D search exactly the same candidates and perform the same table work. The allocation was fixed before reading query results.

7. What the Natural-data Allocator Chose

32-byte vectors: 4 bits per pair

Every A group selected 4×4. Therefore A and D were literally the same representation and returned identical rankings.

64-byte vectors: 8 bits per pair

Only 10–24 of 64 groups differed from 16×16. Almost all were 15×17 or 17×15, using 255 of 256 paid states.

Early mechanism diagnosis

The reconstruction-based allocator found no large unused-cardinality problem on SIFT/GIST. At 64 bytes it mostly moved only one level from one coordinate to its partner.

8. Natural-data Recall Result

At the same number of searched cells, A-minus-D Recall@100 was:

Dataset	coarse cells	minimum	maximum
GIST1M	1,024	-0.00096	+0.00054
GIST1M	4,096	-0.00106	+0.00017
SIFT1M	1,024	-0.000097	+0.000066
SIFT1M	4,096	-0.000033	+0.000436

Meaning in plain language

The largest gain was 0.054 percentage points; the largest loss was 0.106 percentage points. The sign changed across datasets, index sizes, and search ranges, so there is no stable quality improvement.

9. Speed and Baseline Result

- A/D throughput ratio: 0.9918 to 1.0069; group medians are essentially 1.
- A/D p95 latency ratio: 0.9897 to 1.0251.
- This is expected: A and D deliberately execute the same complete-table consumer and nearly the same representation.
- PQ and OPQ generally reach higher Recall in the useful high-quality region.

Natural-data decision

The fixed-adjacent mixed-radix method did not move a useful Recall–speed frontier. A tiny nonzero difference is not a database-systems contribution.

Complete matrix: 512 accepted passes; 32.36 CPU-hours; 17.09 wall-hours; all eight correctness tests pass.

10. Why Add a Synthetic Positive Control?

The natural result left two different explanations:

1. the mixed-radix mechanism is valid, but the required data structure is absent or weak; or
2. the implementation cannot transmit a radix advantage into query ranking, even when such an advantage exists.

Decisive construction

Create exact Cartesian prototypes with the cardinalities required by the intuition. Put 100 identical database vectors at every prototype, place one query exactly on each prototype, search every candidate, and ask whether all 100 true duplicates are recovered.

Positive cases are 3×5 in 4 bits and 15×17 in 8 bits. Matching 4×4 and 16×16 cases are zero-effect controls.

11. Synthetic Result: The Mechanism Reaches Recall

Case	A shape	D shape	code collisions A/D	Recall A/D
positive B4	3×5	4×4	0 / 3	**1.000 / 0.800**
null B4	4×4	4×4	0 / 0	1.000 / 1.000
positive B8	15×17	16×16	0 / 15	**1.000 / 0.941**
null B8	16×16	16×16	0 / 0	1.000 / 1.000

Validity checks

In both null controls A and D produced identical rankings. In both positive cases the rankings differed exactly where D merged prototypes. Two complete runs were byte-for-byte identical.

12. Put the Two Results Together

Question	Natural SIFT/GIST	Synthetic witness
Can A choose non-dyadic radices?	rarely, and only weakly at 64 bytes	yes, exactly 3×5 and 15×17
Does this avoid collisions?	no material measured benefit	yes, 3 and 15 control collisions removed
Does it improve Recall?	tiny, mixed-sign changes	yes, by 0.200 and 0.0588
Does it beat PQ/OPQ?	no useful frontier movement	not tested or claimed

Combined conclusion

The mechanism is real but its necessary structure is not sufficiently strong in the tested natural representation. The synthetic result explains the negative result; it does not reverse it.

13. Coordinate Regrouping Was Not the Missing Contribution

Allowing non-adjacent coordinate pairs improved Recall substantially over the fixed pairing $(0, 1), (2, 3), \ldots$. We then compared our minimum-cost perfect matching with OPQ's existing variance-based grouping rule.

	search 64 cells	search 1,024 cells
SIFT1M: matching minus OPQ grouping	+0.000253	+0.000349
GIST1M: matching minus OPQ grouping	+0.000320	-0.000160

Interpretation

"Do not pair coordinates mechanically" is useful, but already explained by known transform-coding practice. Arbitrary integer radices still add at most about 0.00057 Recall over the identical power-of-two pairing.

14. Why Reconstruction Error Was Not Enough

Recall depends on the ordering of complete estimated distances, not on the average reconstruction error of one coordinate or segment.

The diagnostic question

For each database target, replace one stored segment contribution by its exact distance while leaving every other estimated contribution unchanged. How many wrong pairwise orderings disappear?

- This is an **oracle upper bound**: it uses information that a deployable compressed code does not store.
- A single segment repaired only about 43–44% of inversions and missed the frozen absolute-improvement rule.
- Replacing the 192-dimensional, 6-bit segment together with the 320-dimensional, 4-bit segment repaired **71.1–75.6%**.

What this did and did not establish

Two segments jointly contain ranking headroom. Exact replacement is not an encoder, and the nonlinear Recall ordering test can make independent errors appear jointly important through cancellation.

15. Can a Query-unaware Joint Encoder Use That Headroom?

After the SAQ plan is fixed, the two segments' legal codes form a Cartesian product.
Reconstruction error and unseen-direction worst-case error separate into one objective per segment.
Coupling requires a chosen direction distribution.

Illustrative cancellation

On training directions, two segment errors might be (+2) and (-2), so the total error is zero. On different directions they might become (+2) and (+2), so the same jointly selected pair has total error four. A cancellation seen during training need not survive new directions.

Falsifiable hypothesis

Choose the two codes jointly using one set of base-derived directions, freeze the choices, and evaluate on a disjoint set. Require at least 5% lower MSE than independent selection in both directions and improvement on at least 60% of targets.

16. Frozen Base-only Cross-fit Test

No benchmark query or ground truth was read.

```
for each of 32 held-out database targets:
  alternatives = stored code + every legal one-coordinate +/-1 move
  train on base directions 0..127:
    IND   = best code for each segment separately
    JOINT = best pair over the Cartesian product
  freeze both choices; evaluate on directions 128..255
repeat with the two direction folds swapped
```

Both arms use the same alternatives, bytes, analytical rescale, and estimator. The evaluation oracle chooses after seeing the held-out directions and is reported only as a ceiling, never as a method.

17. Joint Selection Did Not Generalize

	train fold 0	train fold 1
joint MSE change vs independent	+0.0805%	**-1.3644%**
targets improved	15/32	13/32
joint and independent choices differ	31/32	29/32
held-out oracle improvement	16.83%	15.92%

Positive MSE change means an improvement; the negative value means the joint choice is worse.

Decision: JOINT_LOCAL_NO_GO

Good code pairs exist after seeing evaluation directions, but a pair learned from disjoint base directions does not predict them. The observed headroom is direction-specific cancellation, not a stable query-unaware rule.

18. Final Claim Boundary

What remains valid

Mixed-radix packing can preserve non-power-of-two Cartesian cardinalities and prevent collisions in data deliberately having that structure.

What the natural-data evidence rejects

Neither fixed adjacent mixed-radix, optimized coordinate pairing, nor the tested base-trained two-segment joint objective produces a stable new Recall–speed method beyond the closest controls.

Research decision

Close this direct line. Do not enlarge the local code neighbourhood, tune the folds or threshold, or inspect benchmark queries to rescue it. Continuing would require a genuinely new question—for example changing the estimator or storing cross-segment information—followed by a new closest-work and cost review.

References and Evidence

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